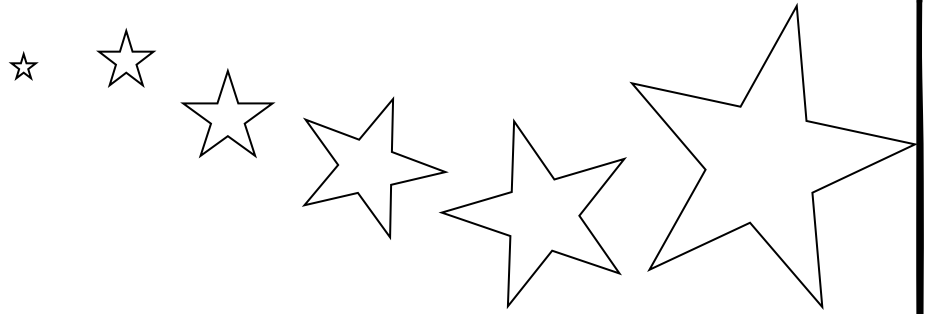
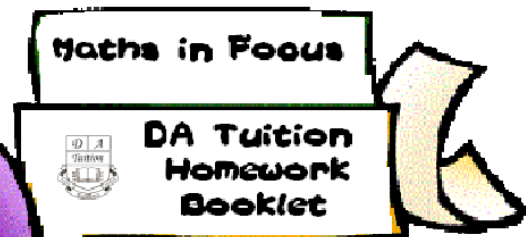
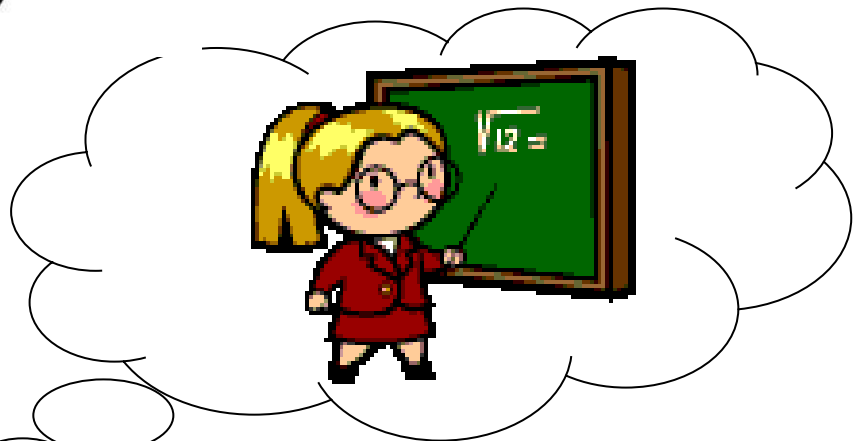


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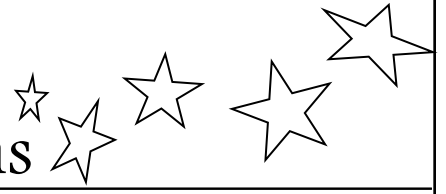


3 UNIT BOS INDUCTION



Name:

S o l u t i o n s



Question 1



2000

Use mathematical induction to prove that

3

$$1 + 3 + 6 + \dots + \frac{1}{2}n(n+1) = \frac{1}{6}n(n+1)(n+2)$$

for all integers $n = 1, 2, 3, \dots$

For $n = 1$, $1 = \frac{1}{6}1(2)(3)$

$$= \frac{1}{6} \times 6$$

$$= 1$$

$\therefore$ The statement is true for $n = 1$.

Assume the statement is true for $n = k$:

i.e. $1 + 3 + 6 + \dots + \frac{1}{2}k(k+1) = \frac{1}{6}k(k+1)(k+2)$

Show that the statement is true for $n = k + 1$; i.e. Show that:

i.e. $1 + 3 + 6 + \dots + \frac{1}{2}k(k+1) + \frac{1}{2}(k+1)(k+2) = \frac{1}{6}(k+1)(k+2)(k+3)$

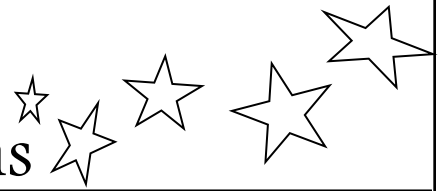
Now $1 + 3 + 6 + \dots + \frac{1}{2}k(k+1) + \frac{1}{2}(k+1)(k+2) = \frac{1}{6}k(k+1)(k+2) + \frac{1}{2}(k+1)(k+2)$

$$= \frac{1}{2}(k+1)(k+2)\left[\frac{1}{3}k+1\right]$$

$$= \frac{1}{2}(k+1)(k+2)\left[\frac{1}{3}(k+3)\right]$$

$$= \frac{1}{6}(k+1)(k+2)(k+3).$$

$\therefore$ If the statement is true for $n = k$, it is true for $n = k + 1$. Hence the statement is true for all integers $n = 1, 2, 3 \dots$ by Mathematical Induction.



Question 2



YEAR 2001

Prove by induction that

3

$$n^3 + (n+1)^3 + (n+2)^3$$

is divisible by 9 for $n = 1, 2, 3, \dots$

To prove:

$$n^3 + (n+1)^3 + (n+2)^3 = 9M$$

for some integer M for $n = 1, 2, 3, \dots$

$$\text{For } n = 1, 1^3 + 2^3 + 3^3 = 36$$

$$= 9 \times 4$$

$\therefore$ The statement is true for $n = 1$.

Assume the statement is true for $n = k$

$$\text{i.e. } k^3 + (k+1)^3 + (k+2)^3 = 9P$$

(for some integer P)

$$\therefore (k+1)^3 + (k+2)^3 = 9P - k^3 \quad \dots (1)$$

Show that the statement is true for $n = k + 1$

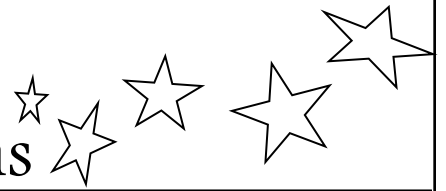
$$\text{i.e. Show that } (k+1)^3 + (k+2)^3 + (k+3)^3 = 9Q$$

(for some integer Q)

$$\begin{aligned} \text{Now } & (k+1)^3 + (k+2)^3 + (k+3)^3 \\ &= 9P - k^3 + (k+3)^3 \quad \text{from (1)} \\ &= 9P - k^3 + k^3 + 9k^2 + 27k + 27 \\ &= 9P + 9k^2 + 27k + 27 \\ &= 9(P + k^2 + 3k + 3) \\ &= 9Q \end{aligned}$$

where Q is the integer $(P + k^2 + 3k + 3)$

$\therefore n^3 + (n+1)^3 + (n+2)^3$ is divisible by 9
for $n = 1, 2, 3, \dots$ by mathematical induction.



Question 3



2002

Use the principle of mathematical induction to show that

3

$$2 \times 1! + 5 \times 2! + 10 \times 3! + \dots + (n^2 + 1)n! = n(n + 1)!$$

for all positive integers n .

To prove:

$$2 \times 1! + 5 \times 2! + 10 \times 3! + \dots + (n^2 + 1)n! = n(n + 1)!$$

for all positive integers n .

$$\text{For } n = 1, \quad \text{LHS} = 2 \times 1! = 2$$

$$\text{RHS} = 1(1 + 1)!$$

$$= 2$$

$\therefore$ The statement is true for $n = 1$.

Assume that the statement is true for $n = k$:

$$\text{i.e. } 2 \times 1! + 5 \times 2! + \dots + (k^2 + 1)k! = k(k + 1)!$$

Show that the statement is true for $n = k + 1$:

$$\text{i.e. } 2 \times 1! + 5 \times 2! + \dots + (k^2 + 1)k! + [(k + 1)^2 + 1](k + 1)! = (k + 1)(k + 1 + 1)! \\ = (k + 1)(k + 2)!$$

$$\text{Now } 2 \times 1! + 5 \times 2! + \dots + (k^2 + 1)k! + [(k + 1)^2 + 1](k + 1)!$$

$$= k(k + 1)! + [(k + 1)^2 + 1](k + 1)!$$

$$= (k + 1)! [k + (k + 1)^2 + 1] \quad (\text{on factorising})$$

$$= (k + 1)! [k + k^2 + 2k + 1 + 1]$$

$$= (k + 1)! (k^2 + 3k + 2)$$

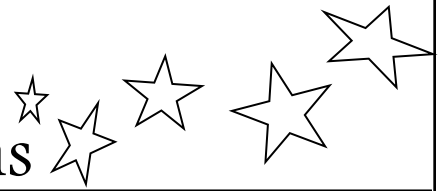
$$= (k + 1)! (k + 2)(k + 1)$$

$$= (k + 2)! (k + 1) \quad (\text{since } (k + 1)!(k + 2) = (k + 2)!) \\ = (k + 1)(k + 2)!$$

Therefore, if the statement is true for $n = k$, it is also true for $n = k + 1$.

And if the statement is true for $n = 1$, it is also true for $n = 2, 3 \dots$ etc.

Therefore the statement is true for all positive integers n , by mathematical induction.



Question 4



2003

3

Use mathematical induction to prove that

$$\frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \frac{1}{5 \times 7} + \dots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1}$$

for all positive integers n .

To prove:

$$\frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \frac{1}{5 \times 7} + \dots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1} \text{ for all positive integers } n.$$

For $n = 1$,

$$\text{LHS} = \frac{1}{1 \times 3} = \frac{1}{3}$$

$$\text{RHS} = \frac{1}{2(1)+1} = \frac{1}{3}$$

$\therefore$ The statement is true for $n = 1$.

Assume that the statement is true for $n = k$:

$$\text{i.e. } \frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \dots + \frac{1}{(2k-1)(2k+1)} = \frac{k}{2k+1}$$

Show that the statement is true for $n = k + 1$:

$$\text{i.e. } \frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \dots + \frac{1}{(2k-1)(2k+1)} + \frac{1}{[2(k+1)-1][2(k+1)+1]} = \frac{k+1}{2(k+1)+1}$$

$$\text{i.e. } \frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \dots + \frac{1}{(2k-1)(2k+1)} + \frac{1}{(2k+1)(2k+3)} = \frac{k+1}{2k+3}$$

$$\text{Now } \frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \dots + \frac{1}{(2k-1)(2k+1)} + \frac{1}{(2k+1)(2k+3)}$$

$$= \frac{k}{2k+1} + \frac{1}{(2k+1)(2k+3)} \quad (\text{using the assumption above})$$

$$= \frac{k(2k+3)+1}{(2k+1)(2k+3)}$$

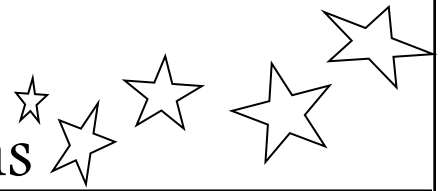
$$= \frac{2k^2+3k+1}{(2k+1)(2k+3)}$$

$$= \frac{(2k+1)(k+1)}{(2k+1)(2k+3)}$$

$$= \frac{k+1}{2k+3}$$

Therefore, if the statement is true for $n = k$, it is also true for $n = k + 1$.

Therefore the statement is true for all positive integers n , by mathematical induction.



Question 5



2004

Use mathematical induction to prove that for all integers $n \geq 3$,

3

$$\left(1 - \frac{2}{3}\right)\left(1 - \frac{2}{4}\right)\left(1 - \frac{2}{5}\right) \dots \left(1 - \frac{2}{n}\right) = \frac{2}{n(n-1)}.$$

Prove the statement is true for $n = 3$:

$$\text{LHS} = 1 - \frac{2}{3}$$

$$= \frac{1}{3}.$$

$$\text{RHS} = \frac{2}{3(3-1)}$$

$$= \frac{2}{6}$$

$$= \frac{1}{3}.$$

$\therefore$ The statement is true for $n = 3$.

Assume that the statement is true for $n = k$:

$$\text{i.e. } \left(1 - \frac{2}{3}\right)\left(1 - \frac{2}{4}\right)\left(1 - \frac{2}{5}\right) \dots \left(1 - \frac{2}{k}\right) = \frac{2}{k(k-1)}$$

Prove that the statement is true for $n = k + 1$:

$$\begin{aligned} \text{LHS} &= \left(1 - \frac{2}{3}\right)\left(1 - \frac{2}{4}\right) \dots \left(1 - \frac{2}{k}\right)\left(1 - \frac{2}{k+1}\right) \\ &= \left(\frac{2}{k(k-1)}\right)\left(1 - \frac{2}{k+1}\right) \text{ (using assumption)} \end{aligned}$$

$$= \frac{2}{k(k-1)} - \frac{4}{k(k-1)(k+1)}$$

$$= \frac{2(k+1) - 4}{k(k-1)(k+1)}$$

$$= \frac{2k + 2 - 4}{k(k-1)(k+1)}$$

$$= \frac{2k - 2}{k(k-1)(k+1)}$$

$$= \frac{2(k-1)}{k(k-1)(k+1)}$$

$$= \frac{2}{k(k+1)}$$

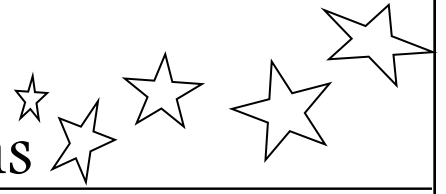
$$\text{RHS} = \frac{2}{(k+1)(k+1-1)}$$

$$= \frac{2}{k(k+1)}$$

$$= \text{LHS}$$

$\therefore$ The statement is true for $n = k + 1$.

$\therefore$ If the statement is true for $n = 3$ and $n = k + 1$ then it is true for $n = 4, 5$ and so on for all positive integers $n \geq 3$.



Question 6



2008

Use mathematical induction to prove that, for integers $n \geq 1$,

3

$$1 \times 3 + 2 \times 4 + 3 \times 5 + \dots + n(n+2) = \frac{n}{6}(n+1)(2n+7).$$

Q3b When $n = 1$, $\frac{1}{6}(1+1)(2(1)+7) = 3 = 1 \times 3$, the statement is

true. Assume that it is true when $n = k$,

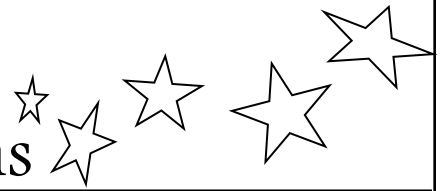
i.e. $1 \times 3 + 2 \times 4 + 3 \times 5 + \dots + k(k+2) = \frac{k}{6}(k+1)(2k+7)$, then

$$1 \times 3 + \dots + k(k+2) + (k+1)(k+3) = \frac{k}{6}(k+1)(2k+7) + (k+1)(k+3)$$

$$= (k+1) \left(\frac{2k^2 + 13k + 18}{6} \right) = \frac{(k+1)}{6} (k+2)(2k+9)$$

$$= \frac{(k+1)}{6} ((k+1)+1)(2(k+1)+7). \therefore \text{it is also true for } n = k+1.$$

Hence it is true for all $n \geq 1$.



Question 7



2010

3

Prove by induction that

$$47^n + 53 \times 147^{n-1}$$

is divisible by 100 for all integers $n \geq 1$.

Let $n = 1$, $47 + 53 \times 147^0 = 47 + 53 = 100$, which is divisible by 100.

Assume $47^n + 53 \times 147^{n-1}$ is divisible by 100,

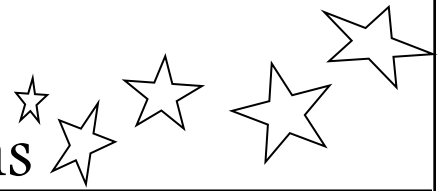
let $47^n + 53 \times 147^{n-1} = 100M$, where M is an integer.

$$\therefore 47^n = 100M - 53 \times 147^{n-1}.$$

$$\begin{aligned} 47^{n+1} + 53 \times 147^n &= 47(100M - 53 \times 147^{n-1}) + 53 \times 147^n \\ &= 4700M + 53 \times 147^{n-1}(-47 + 147) \\ &= 4700M + 5300 \times 147^{n-1} \\ &= 100(47M + 53 \times 147^{n-1}). \end{aligned}$$

$\therefore 47^{n+1} + 53 \times 147^n$ is divisible by 100.

By the Principle of Induction, $47^n + 53 \times 147^{n-1}$ is divisible by 100 for all $n > 1$.



Question 8



2011

3

Use mathematical induction to prove that, for $n \geq 1$,

$$1 \times 5 + 2 \times 6 + 3 \times 7 + \dots + n(n+4) = \frac{1}{6}n(n+1)(2n+13).$$

(i) Let $n=1$, LHS $= 1 \times 5 = 5$,

$$\text{RHS} = \frac{1}{6} \times 1 \times 2 \times 15 = 5 \therefore \text{True.}$$

Assume $1 \times 5 + 2 \times 6 + \dots + n(n+4)$

$$= \frac{1}{6}n(n+1)(2n+13).$$

RTP $1 \times 5 + 2 \times 6 + \dots + (n+1)(n+5)$

$$= \frac{1}{6}(n+1)(n+2)(2n+15).$$

$$\text{LHS} = \frac{1}{6}n(n+1)(2n+13) + (n+1)(n+5)$$

$$= \frac{1}{6}(n+1)(n(2n+13) + 6(n+5))$$

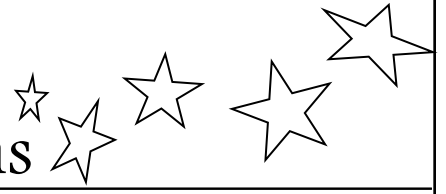
$$= \frac{1}{6}(n+1)(2n^2 + 13n + 6n + 30)$$

$$= \frac{1}{6}(n+1)(2n^2 + 19n + 30)$$

$$= \frac{1}{6}(n+1)(n+2)(2n+15) = \text{RHS.}$$

$\therefore$ True for $n+1$.

$\therefore$ True for all $n \geq 1$ by the principle of Induction.



Question 9



2012

Use mathematical induction to prove that $2^{3n} - 3^n$ is divisible by 5 for $n \geq 1$.

(a) Let $n = 1$, $2^3 - 3 = 8 - 3 = 5$, hence, divisible by 5.

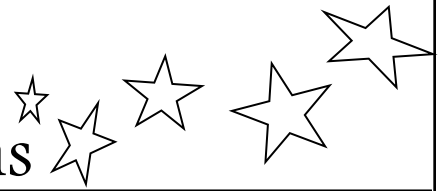
Assume $2^{3n} - 3^n = 5M$, where M is an integer.

$$\therefore 2^{3n} = 5M + 3^n.$$

Required to prove that $2^{3(n+1)} - 3^{n+1}$ is divisible by 5.

$$\begin{aligned} 2^{3(n+1)} - 3^{n+1} &= 2^3 \times 2^{3n} - 3 \times 3^n \\ &= 8(5M + 3^n) - 3 \times 3^n \\ &= 40M + 5 \times 3^n \\ &= 5(8M + 3^n), \text{ which is divisible by 5.} \end{aligned}$$

$\therefore$ By the principle of Induction, $2^{3n} - 3^n$ is divisible by 5 for all $n \geq 1$.



Question 10



2014

Use mathematical induction to prove that $2^n + (-1)^{n+1}$ is divisible by 3 for all integers $n \geq 1$.

3

$$2^n + (-1)^{n+1}$$

$$n = 1, 2^1 + (-1)^2 = 2 + 1 = 3 \text{ which is divisible by 3}$$

Assume true for $n = k$

ie Assume $2^k + (-1)^{k+1}$ is divisible by 3

ie $2^k + (-1)^{k+1} = 3M$ where M is an integer.

We need to prove true for $n = k + 1$

ie That $2^{k+1} + (-1)^{k+2}$ is divisible by 3

$$\begin{aligned} 2^{k+1} + (-1)^{k+2} &= 2(2^k) + (-1)^{k+2} \\ &= 2[3M + (-1)^{k+1}] + (-1)^{k+2} \\ &= 3[2M + (-1)^{k+1}] \end{aligned} \quad \left| \begin{aligned} 2^k &= 3M - (-1)^{k+1} \\ &= 3M + (-1)^{k+2} \end{aligned} \right.$$

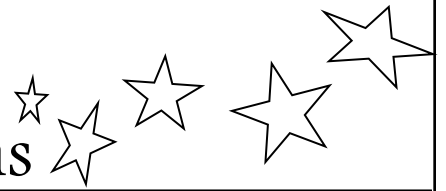
Which is a multiple of 3 since M and k are integers.

ie if $2^k + (-1)^{k+1}$ is divisible by 3, then

$2^{k+1} + (-1)^{k+2}$ is divisible by 3

$\therefore$ By the principle of mathematical induction

$2^n + (-1)^{n+1}$ is divisible by 3 for all $n \geq 1$.



Question 11



2015

Prove by mathematical induction that for all integers $n \geq 1$,

3

$$\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{n}{(n+1)!} = 1 - \frac{1}{(n+1)!}.$$

$$\begin{aligned} \text{When } n = 1, \quad RHS &= 1 - \frac{1}{(1+1)!} \\ &= 1 - \frac{1}{2!} = \frac{1}{2} = LHS \end{aligned}$$

Statement true for $n = 1$.

Assume the result is true for $n = k$

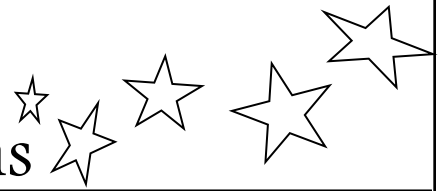
ie assume $\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{k}{(k+1)!} = 1 - \frac{1}{(k+1)!}$ to be true.

Prove true for $n = k + 1$

$$\text{ie } \frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{k}{(k+1)!} + \frac{k+1}{(k+1+1)!} = 1 - \frac{1}{(k+1+1)!}$$

$$\begin{aligned} LHS &= 1 - \frac{1}{(k+1)!} + \frac{k+1}{(k+2)!} && \text{(from assumption)} \\ &= 1 - \frac{(k+2) - (k+1)}{(k+2)!} \\ &= 1 - \frac{1}{(k+2)!} = RHS \end{aligned}$$

Hence the result is proven by mathematical induction.



Question 12



2016

(i) Show that $4n^3 + 18n^2 + 23n + 9$ can be written as

1

$$(n+1)(4n^2 + 14n + 9).$$

(ii) Using the result in part (i), or otherwise, prove by mathematical induction that, for $n \geq 1$,

3

$$1 \times 3 + 3 \times 5 + 5 \times 7 + \dots + (2n-1)(2n+1) = \frac{1}{3}n(4n^2 + 6n - 1).$$

R.T.P. $4n^3 + 18n^2 + 23n + 9$

$$= (n+1)(4n^2 + 14n + 9)$$

$$\text{RHS} = n(4n^2 + 14n + 9) + 1(4n^2 + 14n + 9)$$

$$= 4n^3 + 14n^2 + 9n + 4n^2 + 14n + 9$$

$$= 4n^3 + 18n^2 + 23n + 9$$

$$= \text{LHS}$$

OR

$$\text{LHS} = 4n^3 + 4n^2 + 14n^2 + 14n + 9n + 9$$

$$= 4n^2(n+1) + 14n(n+1) + 9(n+1)$$

$$= (n+1)(4n^2 + 14n + 9)$$

$$= \text{RHS}$$

Let $T_n = (2n-1)(2n+1)$, $S_n = \frac{1}{3}n(4n^2 + 6n - 1)$

R.T.P. $1 \times 3 + 3 \times 5 + 5 \times 7 + \dots + (2n-1)(2n+1) = \frac{1}{3}n(4n^2 + 6n - 1)$

When $n = 1$, $\text{LHS} = 1 \times 3$ $\text{RHS} = \frac{1}{3}(1)(4(1)^2 + 6(1) - 1)$

$$= 3$$

$$= \frac{1}{3} \times 9$$

$$= 3$$

$\therefore$ True for $n=1$

Assume true for some integer k , ie $S_k = \frac{1}{3}k(4k^2 + 6k - 1)$

and attempt to show it is also true for $k+1$

ie, try to show $S_{k+1} = \frac{1}{3}(k+1)(4(k+1)^2 + 6(k+1) - 1)$

$$= \frac{1}{3}(k+1)(4k^2 + 8k + 4 + 6k + 6 - 1)$$

$$= \frac{1}{3}(k+1)(4k^2 + 14k + 9)$$

Now $S_{k+1} = S_k + T_{k+1}$

$$= \frac{1}{3}k(4k^2 + 6k - 1) + (2(k+1)-1)(2(k+1)+1)$$

$$= \frac{1}{3}k(4k^2 + 6k - 1) + (2k+1)(2k+3)$$

$$= \frac{1}{3}(4k^3 + 6k^2 - k) + 4k^2 + 8k + 3$$

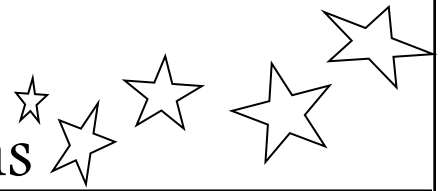
$$= \frac{1}{3}(4k^3 + 6k^2 - k + 12k^2 + 24k + 9)$$

$$= \frac{1}{3}(4k^3 + 18k^2 + 23k + 9)$$

$$= \frac{1}{3}(k+1)(4k^2 + 14k + 9) \quad (\text{From part (i)})$$

as required

$\therefore$ Hence the result is true for all positive integers n , by induction.



Question 13



2017

Prove by mathematical induction that $8^{2n+1} + 6^{2n-1}$ is divisible by 7, for any integer $n \geq 1$.

3

Let $P(n)$ be the given proposition.

$P(1)$ is true since $8^3 + 6 = 518 = 7 \times 74$ which is divisible by 7.

Let k be an integer for which $P(k)$ is true.

That is $8^{2k+1} + 6^{2k-1} = 7m$, for some integer m .

Consider $8^{2(k+1)+1} + 6^{2(k+1)-1}$

$$= 8^2 \times 8^{2k+1} + 6^2 \times 6^{2k-1}$$

$$= 8^2 \times 8^{2k+1} + 8^2 \times 6^{2k-1} - 8^2 \times 6^{2k-1} + 6^2 \times 6^{2k-1}$$

$$= 8^2 (8^{2k+1} + 6^{2k-1}) + (6^2 - 8^2) 6^{2k-1}$$

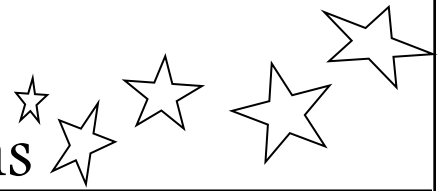
$$= 64 \times 7m - 28 \times 6^{2k-1}$$

$$= 7(64m - 4 \times 6^{2k-1})$$

which is divisible by 7, since m and k are integers and $64m - 4 \times 6^{2k-1}$ is an integer.

$\therefore P(k+1)$ is also true

$\therefore P(n)$ is true for all $n \geq 1$ by induction.



Question 14



2018

Prove by mathematical induction that, for $n \geq 1$,

3

$$2 - 6 + 18 - 54 + \dots + 2(-3)^{n-1} = \frac{1 - (-3)^n}{2}.$$

RTP: $2 - 6 + 18 - 54 + \dots + 2(-3)^{n-1} = \frac{1 - (-3)^n}{2}$

Show true for $n = 1$:

LHS = $2(-3)^{1-1}$

= 2×1

= 2

RHS = $\frac{1 - (-3)^1}{2}$

= $\frac{1 + 3}{2}$

= 2

LHS = RHS $\therefore$ true for $n = 1$

Assume true for $n = k$:

$$2 - 6 + 18 - 54 + \dots + 2(-3)^{k-1} = \frac{1 - (-3)^k}{2}$$

If $n = k$ is true, show true for $n = k + 1$.

LHS = $2 - 6 + 18 - 54 + \dots + 2(-3)^{k-1} + 2(-3)^k$

= $\frac{1 - (-3)^k}{2} + 2(-3)^k$

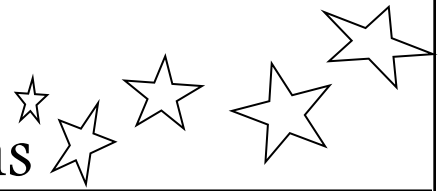
= $\frac{1 - (-3)^k + 4(-3)^k}{2}$

= $\frac{1 - (-3)(-3)^k}{2}$

= $\frac{1 - (-3)^{k+1}}{2}$

$\therefore$ true for $n = k + 1$.

Hence, by the process of mathematical induction $2 - 6 + 18 - 54 + \dots + 2(-3)^{n-1} = \frac{1 - (-3)^n}{2}$.



Question 15



2019

Prove by mathematical induction that, for all integers $n \geq 1$,

3

$$1(1!) + 2(2!) + 3(3!) + \cdots + n(n!) = (n + 1)! - 1.$$

$$\begin{aligned} n = 1: \quad \text{LHS} &= 1(1!) = 1 \\ \text{RHS} &= (1 + 1)! - 1 = 1 = \text{LHS} \end{aligned}$$

True when $n = 1$

Suppose true for $n = k$

$$\text{So } 1(1!) + 2(2!) + \cdots + k(k!) = (k + 1)! - 1$$

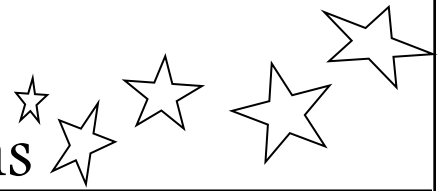
Want to show it is then true for $n = k + 1$

$$\text{that is } 1(1!) + \cdots + k(k!) + (k + 1)((k + 1)!) = ((k + 1) + 1)! - 1$$

$$\begin{aligned} \text{LHS} &= 1(1!) + \cdots + k(k!) + (k + 1)((k + 1)!) \\ &= ((k + 1)! - 1) + (k + 1)((k + 1)!) \\ &= (k + 1)! + (k + 1)(k + 1)! - 1 \\ &= (k + 1)!(1 + k + 1) - 1 \\ &= (k + 1)!(k + 2) - 1 \\ &= (k + 2)! - 1 \\ &= \text{RHS} \end{aligned}$$

If true for $n = k$, then true for $n = k + 1$

Hence, by mathematical induction, true for $n \geq 1$.



Question 16



2020

Use the principle of mathematical induction to show that for all integers $n \geq 1$, 3

$$1 \times 2 + 2 \times 5 + 3 \times 8 + \cdots + n(3n - 1) = n^2(n + 1).$$

$$(1 \times 2) + (2 \times 5) + (3 \times 8) + \cdots + n(3n - 1) = n^2(n + 1)$$

For $n = 1$	LHS = 1 (2)	RHS = 1 (2)
	= 2	= 2

$\therefore$ Statement is true for $n = 1$

Assume statement true for $n = k$, that is,

$$(1 \times 2) + (2 \times 5) + (3 \times 8) + \cdots + k(3k - 1) = k^2(k + 1)$$

Prove true for $n = k + 1$

That is, we show that

$$(1 \times 2) + (2 \times 5) + \cdots + k(3k - 1) + (k + 1)(3(k + 1) - 1) = (k + 1)^2(k + 2)$$

$$\text{LHS} = k^2(k + 1) + (k + 1)(3k + 2)$$

$$= (k + 1)(k^2 + 3k + 2)$$

$$= (k + 1)(k + 1)(k + 2)$$

$$= (k + 1)^2(k + 2)$$

$$= \text{RHS}$$

$\therefore$ By the principle of mathematical induction the statement is true for $n \geq 1$.